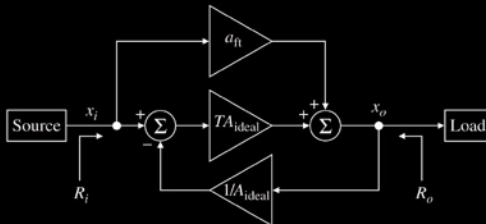


Fourth Edition

Design with Operational Amplifiers and Analog Integrated Circuits



Sergio Franco

DESIGN WITH OPERATIONAL AMPLIFIERS AND ANALOG INTEGRATED CIRCUITS

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DESIGN WITH OPERATIONAL AMPLIFIERS AND ANALOG INTEGRATED CIRCUITS

FOURTH EDITION

Sergio Franco

San Francisco State University





DESIGN WITH OPERATIONAL AMPLIFIERS AND ANALOG INTEGRATED CIRCUITS, FOURTH EDITION

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Sergio Franco was born in Friuli, Italy, and earned his Ph.D. from the University of Illinois at Urbana-Champaign. After working in industry, both in the United States and Italy, he joined San Francisco State University in 1980, where he has contributed to the formation of many hundreds of successful analog engineers gainfully employed in Silicon Valley. Dr. Franco is the author of the textbook *Analog Circuit Design—Discrete & Integrated*, also by McGraw-Hill. More information can be found in the author's website at <http://online.sfsu.edu/sfranco/>.

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PREFACE

During the last decades much has been prophesized that there will be little need for analog circuitry in the future because digital electronics is taking over. Far from having proven true, this contention has provoked controversial rebuttals, as epitomized by statements such as “If you cannot do it in digital, it’s got to be done in analog.” Add to this the common misconception that analog design, compared to digital design, seems to be more of a whimsical art than a systematic science, and what is the confused student to make of this controversy? Is it worth pursuing some coursework in analog electronics, or is it better to focus just on digital?

There is no doubt that many functions that were traditionally the domain of analog electronics are nowadays implemented in digital form, a popular example being offered by digital audio. Here, the analog signals produced by microphones and other acoustic transducers are suitably conditioned by means of amplifiers and filters, and are then converted to digital form for further processing, such as mixing, editing, and the creation of special effects, as well as for the more mundane but no less important tasks of transmission, storage, and retrieval. Finally, digital information is converted back to analog signals for playing through loudspeakers. One of the main reasons why it is desirable to perform as many functions as possible digitally is the generally superior reliability and flexibility of digital circuitry. However, *the physical world is inherently analog*, indicating that there will *always* be a need for analog circuitry to condition physical signals such as those associated with transducers, as well as to convert information from analog to digital for processing, and from digital back to analog for reuse in the physical world. Moreover, new applications continue to emerge, where considerations of speed and power make it more advantageous to use analog front ends; wireless communications provide a good example.

Indeed many applications today are best addressed by mixed-mode integrated circuits (mixed-mode ICs) and systems, which rely on analog circuitry to interface with the physical world, and digital circuitry for processing and control. Even though the analog circuitry may constitute only a small portion of the total chip area, it is often the most challenging part to design as well as the limiting factor on the performance of the entire system. In this respect, it is usually the analog designer who is called to devise ingenious solutions to the task of realizing analog functions in decidedly digital technologies; switched-capacitor techniques in filtering and sigma-delta techniques in data conversion are popular examples. In light of the above, the need for competent analog designers will continue to remain very strong. Even purely digital circuits, when pushed to their operational limits, exhibit analog behavior. Consequently, a solid grasp of analog design principles and techniques is a valuable asset in the design of any IC, not just purely digital or purely analog ICs.

THE BOOK

The goal of this book is the illustration of general analog principles and design methodologies using practical devices and applications. The book is intended as a

textbook for undergraduate and graduate courses in design and applications with analog integrated circuits (analog ICs), as well as a reference book for practicing engineers. The reader is expected to have had an introductory course in electronics, to be conversant in frequency-domain analysis techniques, and to possess basic skills in the use of SPICE. Though the book contains enough material for a two-semester course, it can also serve as the basis for a one-semester course after suitable selection of topics. The selection process is facilitated by the fact that the book as well as its individual chapters have generally been designed to proceed from the elementary to the complex.

At San Francisco State University we have been using the book for a sequence of two one-semester courses, one at the senior and the other at the graduate level. In the senior course we cover Chapters 1–3, Chapters 5 and 6, and most of Chapters 9 and 10; in the graduate course we cover all the rest. The senior course is taken concurrently with a course in analog IC fabrication and design. For an effective utilization of analog ICs, it is important that the user be cognizant of their internal workings, at least qualitatively. To serve this need, the book provides intuitive explanations of the technological and circuital factors intervening in a design decision.

NEW TO THE FOURTH EDITION

The key features of the new edition are: (a) a complete revision of negative feedback, (b) much enhanced treatment of op amp dynamics and frequency compensation, (c) expanded coverage of switching regulators, (d) a more balanced presentation of bipolar and CMOS technologies, (e) a substantial increase of in-text PSpice usage, and (f) redesigned examples and about 25% new end-of-chapter problems to reflect the revisions.

While previous editions addressed negative feedback from the specialized viewpoint of the op amp user, the fourth edition offers a much broader perspective that will prove useful also in other areas like switching regulators and phase-locked loops. The new edition presents both two-port analysis and return-ratio analysis, emphasizing similarities but also differences, in an attempt at dispelling the persisting confusion between the two (to keep the distinction, the loop gain and the feedback factor are denoted as L and b in two-port analysis, and as T and β in return-ratio analysis).

Of necessity, the feedback revision is accompanied by an extensive rewriting of op amp dynamics and frequency compensation. In this connection, the fourth edition makes generous use of the voltage/current injection techniques pioneered by R. D. Middlebrook for loop-gain measurements.

In view of the importance of portable-power management in today's analog electronics, this edition offers an expanded coverage of switching regulators. Much greater attention is devoted to current control and slope compensation, along with stability issues such as the effect of the right-half plane zero and error-amplifier design.

The book makes abundant use of SPICE (schematic capture instead of the netlists of the previous editions), both to verify calculations and to investigate higher-order effects that would be too complex for paper and pencil analysis. SPICE is nowadays available in a variety of versions undergoing constant revision, so rather than committing to a particular version, I have decided to keep the examples simple

enough for students to quickly redraw them and run them in the SPICE version of their choice.

As in the previous editions, the presentation is enhanced by carefully thought-out examples and end-of-chapter problems emphasizing intuition, physical insight, and problem-solving methodologies of the type engineers exercise daily on the job.

The desire to address general and lasting principles in a manner that transcends the latest technological trend has motivated the choice of well-established and widely documented devices as vehicles. However, when necessary, students are made aware of more recent alternatives, which they are encouraged to look up online.

THE CONTENTS AT A GLANCE

Although not explicitly indicated, the book consists of three parts. The first part (Chapters 1–4) introduces fundamental concepts and applications based on the op amp as a predominantly ideal device. It is felt that the student needs to develop sufficient confidence with ideal (or near-ideal) op amp situations before tackling and assessing the consequences of practical device limitations. Limitations are the subject of the second part (Chapters 5–8), which covers the topic in more systematic detail than previous editions. Finally, the third part (Chapters 9–13) exploits the maturity and judgment developed by the reader in the first two parts to address a variety of design-oriented applications. Following is a brief chapter-by-chapter description of the material covered.

Chapter 1 reviews basic amplifier concepts, including negative feedback. Much emphasis is placed on the loop gain as a gauge of circuit performance. The loop gain is treated via both two-port analysis and return-ratio analysis, with due attention to similarities as well as differences between the two approaches. The student is introduced to simple PSpice models, which will become more sophisticated as we progress through the book. Those instructors who find the loop-gain treatment overwhelming this early in the book may skip it and return to it at a more suitable time. Coverage rearrangements of this sort are facilitated by the fact that individual sections and chapters have been designed to be as independent as possible from each other; moreover, the end-of-chapter problems are grouped by section.

Chapter 2 deals with I - V , V - I , and I - I converters, along with various instrumentation and transducer amplifiers. The chapter places much emphasis on feedback topologies and the role of the loop gain T .

Chapter 3 covers first-order filters, audio filters, and popular second-order filters such as the KRC , multiple-feedback, state-variable, and biquad topologies. The chapter emphasizes complex-plane systems concepts and concludes with filter sensitivities.

The reader who wants to go deeper into the subject of filters will find Chapter 4 useful. This chapter covers higher-order filter synthesis using both the cascade and the direct approaches. Moreover, these approaches are presented for both the case of active RC filters and the case of switched-capacitor (SC) filters.

Chapter 5 addresses input-referrable op amp errors such as V_{OS} , I_B , I_{OS} , CMRR, PSRR, and drift, along with operating limits. The student is introduced to data-sheet interpretation, PSpice macromodels, and also to different technologies and topologies.

Chapter 6 addresses dynamic limitations in both the frequency and time domains, and investigates their effect on the resistive circuits and the filters that were studied in the first part using mainly ideal op amp models. Voltage feedback and current feedback are compared in detail, and PSpice is used extensively to visualize both the frequency and transient responses of representative circuit examples. Having mastered the material of the first four chapters using ideal or nearly ideal op amps, the student is now in a better position to appreciate and evaluate the consequences of practical device limitations.

The subject of ac noise, covered in Chapter 7, follows naturally since it combines the principles learned in both Chapters 5 and 6. Noise calculations and estimation represent another area in which PSpice proves a most useful tool.

The second part concludes with the subject of stability in Chapter 8. The enhanced coverage of negative feedback has required an extensive revision of frequency compensation, both internal and external to the op amp. The fourth edition makes generous use of the voltage/current injection techniques pioneered by R. D. Middlebrook for loop-gain measurements. Again, PSpice is used profusely to visualize the effect of the different frequency-compensation techniques presented.

The third part begins with nonlinear applications, which are discussed in Chapter 9. Here, nonlinear behavior stems from either the lack of feedback (voltage comparators), or the presence of feedback, but of the positive type (Schmitt triggers), or the presence of negative feedback, but using nonlinear elements such as diodes and switches (precision rectifiers, peak detectors, track-and-hold amplifiers).

Chapter 10 covers signal generators, including Wien-bridge and quadrature oscillators, multivibrators, timers, function generators, and V - F and F - V converters.

Chapter 11 addresses regulation. It starts with voltage references, proceeds to linear voltage regulators, and concludes with a much-expanded coverage of switching regulators. Great attention is devoted to current control and slope compensation, along with stability issues such as error-amplifier design and the effect of the right-half plane zero in boost converters.

Chapter 12 deals with data conversion. Data-converter specifications are treated in systematic fashion, and various applications with multiplying DACs are presented. The chapter concludes with oversampling-conversion principles and sigma-delta converters. Much has been written about this subject, so this chapter of necessity exposes the student only to the fundamentals.

Chapter 13 concludes the book with a variety of nonlinear circuits, such as log/antilog amplifiers, analog multipliers, and operational transconductance amplifiers with a brief exposure to g_m - C filters. The chapter culminates with an introduction to phase-locked loops, a subject that combines important materials addressed at various points in the preceding chapters.

WEBSITE

The book is accompanied by a Website (<http://www.mhhe.com/franco>) containing information about the book and a collection of useful resources for the instructor. Among the Instructor Resources are a Solutions Manual, a set of PowerPoint Lecture Slides, and a link to the Errata.

the cost of a print textbook, reduce their impact on the environment, and gain access to powerful web tools for learning. CourseSmart eBooks can be viewed online or downloaded to a computer. The eBooks allow readers to do full text searches, add highlighting and notes, and share notes with others. CourseSmart has the largest selection of eBooks available anywhere. Visit www.CourseSmart.com to learn more and to try a sample chapter.

ACKNOWLEDGMENTS

Some of the changes in the fourth edition were made in response to feedback received from a number of readers in both industry and academia, and I am grateful to all who took the time to e-mail me. In addition, the following reviewers provided detailed commentaries on the previous edition as well as valuable suggestions for the current revision. All suggestions have been examined in detail, and if only a portion of them has been honored, it was not out of callousness, but because of production constraints or personal philosophy. To all reviewers, my sincere thanks: Aydin Karsilayan, Texas A&M University; Paul T. Kolen, San Diego State University; Jih-Sheng (Jason) Lai, Virginia Tech; Andrew Rusek, Oakland University; Ashok Srivastava, Louisiana State University; S. Yuvarajan, North Dakota State University.

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Sergio Franco
San Francisco, California, 2014

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1

OPERATIONAL AMPLIFIER FUNDAMENTALS

- 1.1 Amplifier Fundamentals
- 1.2 The Operational Amplifier
- 1.3 Basic Op Amp Configurations
- 1.4 Ideal Op Amp Circuit Analysis
- 1.5 Negative Feedback
- 1.6 Feedback in Op Amp Circuits
- 1.7 The Return Ratio and Blackman's Formula
- 1.8 Op Amp Powering
 - Problems
 - References
 - Appendix 1A Standard Resistance Values

The term *operational amplifier*, or *op amp* for short, was coined in 1947 by John R. Ragazzini to denote a special type of amplifier that, by proper selection of its external components, could be configured for a variety of operations such as amplification, addition, subtraction, differentiation, and integration. The first applications of op amps were in analog computers. The ability to perform mathematical operations was the result of combining high gain with negative feedback.

Early op amps were implemented with vacuum tubes, so they were bulky, power-hungry, and expensive. The first dramatic miniaturization of the op amp came with the advent of the bipolar junction transistor (BJT), which led to a whole generation of op amp modules implemented with discrete BJTs. However, the real breakthrough occurred with the development of the integrated circuit (IC) op amp, whose elements are fabricated in monolithic form on a silicon chip the size of a pinhead. The first such device was developed by Robert J. Widlar at Fairchild Semiconductor Corporation in the early 1960s. In 1968 Fairchild introduced the op amp that was to become the industry standard, the popular $\mu A741$. Since then the number of op amp families and manufacturers has swollen considerably. Nevertheless, the 741 is undoubtedly the

most widely documented op amp. Building blocks pioneered by the 741 continue to be in widespread use today, and current literature still refers to classic 741 articles, so it pays to study this device both from a historical perspective and a pedagogical standpoint.

Op amps have made lasting inroads into virtually every area of analog and mixed analog-digital electronics.¹ Such widespread use has been aided by dramatic price drops. Today, the cost of an op amp that is purchased in volume quantities can be comparable to that of more traditional and less sophisticated components such as trimmers, quality capacitors, and precision resistors. In fact, the prevailing attitude is to regard the op amp as just another component, a viewpoint that has had a profound impact on the way we think of analog circuits and design them today.

The internal circuit diagram of the 741 op amp is shown in Fig. 5A.2 of the Appendix at the end of Chapter 5. The circuit may be intimidating, especially if you haven't been exposed to BJTs in sufficient depth. Be reassured, however, that it is possible to design a great number of op amp circuits without a detailed knowledge of the op amp's inner workings. Indeed, in spite of its internal complexity, the op amp lends itself to a black-box representation with a very simple relationship between output and input. We shall see that this simplified schematization is adequate for a great variety of situations. When it is not, we shall turn to the data sheets and predict circuit performance from specified data, again avoiding a detailed consideration of the inner workings.

To promote their products, op amp manufacturers maintain applications departments with the purpose of identifying areas of application for their products and publicizing them by means of application notes and articles in trade magazines. Nowadays much of this information is available on the web, which you are encouraged to browse in your spare time to familiarize yourself with analog-products data sheets and application notes. You can even sign up for online seminars, or "webinars."

This study of op amp principles should be corroborated by practical experimentation. You can either assemble your circuits on a protoboard and try them out in the lab, or you can simulate them with a personal computer using any of the various CAD/CAE packages available, such as SPICE. For best results, you may wish to do both.

Chapter Highlights

After reviewing basic amplifier concepts, the chapter introduces the op amp and presents analytical techniques suitable for investigating a variety of basic op amp circuits such as inverting/non-inverting amplifiers, buffers, summing/difference amplifiers, differentiators/integrators, and negative-resistance converters.

Central to the operation of op amp circuits is the concept of negative feedback, which is investigated next. Both two-port analysis and return-ratio analysis are presented, and with a concerted effort at dispelling notorious confusion between the two approaches. (To differentiate between the two, the loop gain and the feedback factor are denoted as L and b in the two-port approach, and as T and β in the return-ratio approach). The benefits of negative feedback are illustrated with a generous amount of examples and SPICE simulations.

The chapter concludes with practical considerations such as op amp powering, internal power dissipation, and output saturation. (Practical limitations will be taken up again and in far greater detail in Chapters 5 and 6.) The chapter makes abundant use of SPICE, both as a validation tool for hand calculations, and as a pedagogical tool to confer more immediacy to concepts and principles as they are first introduced.

1.1 AMPLIFIER FUNDAMENTALS

Before embarking on the study of the operational amplifier, it is worth reviewing the fundamental concepts of amplification and loading. Recall that an amplifier is a two-port device that accepts an externally applied signal, called *input*, and generates a signal called *output* such that $output = gain \times input$, where *gain* is a suitable proportionality constant. A device conforming to this definition is called a *linear amplifier* to distinguish it from devices with nonlinear input-output relationships, such as quadratic and log/antilog amplifiers. Unless stated to the contrary, the term *amplifier* will here signify *linear amplifier*.

An amplifier receives its input from a *source* upstream and delivers its output to a *load* downstream. Depending on the nature of the input and output signals, we have different amplifier types. The most common is the *voltage amplifier*; whose input v_I and output v_O are voltages. Each port of the amplifier can be modeled with a Thévenin equivalent, consisting of a voltage source and a series resistance. The input port usually plays a purely passive role, so we model it with just a resistance R_i , called the *input resistance* of the amplifier. The output port is modeled with a voltage-controlled voltage source (VCVS) to signify the dependence of v_O on v_I , along with a series resistance R_o called the *output resistance*. The situation is depicted in Fig. 1.1, where A_{oc} is called the *voltage gain factor* and is expressed in volts per volt. Note that the input source is also modeled with a Thévenin equivalent consisting of the source v_S and an internal series resistance R_s ; the output load, playing a passive role, is modeled with a mere resistance R_L .

We now wish to derive an expression for v_O in terms of v_S . Applying the voltage divider formula at the output port yields

$$v_O = \frac{R_L}{R_o + R_L} A_{oc} v_I \quad (1.1)$$

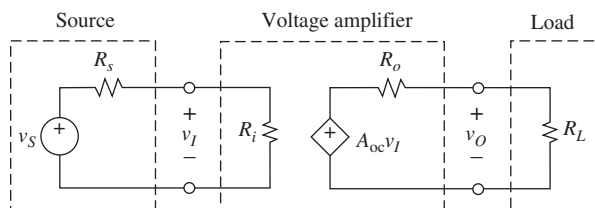


FIGURE 1.1
Voltage amplifier.

We note that in the absence of any load ($R_L = \infty$) we would have $v_O = A_{oc}v_I$. Hence, A_{oc} is called the *unloaded*, or *open-circuit*, voltage gain. Applying the voltage divider formula at the input port yields

$$v_I = \frac{R_i}{R_s + R_i} v_S \quad (1.2)$$

Eliminating v_I and rearranging, we obtain the *source-to-load gain*,

$$\frac{v_O}{v_S} = \frac{R_i}{R_s + R_i} A_{oc} \frac{R_L}{R_o + R_L} \quad (1.3)$$

As the signal progresses from source to load, it undergoes first some attenuation at the input port, then magnification by A_{oc} inside the amplifier, and finally additional attenuation at the output port. These attenuations are referred to as *loading*. It is apparent that because of loading, Eq. (1.3) gives $|v_O/v_S| \leq |A_{oc}|$.

EXAMPLE 1.1. (a) An amplifier with $R_i = 100 \text{ k}\Omega$, $A_{oc} = 100 \text{ V/V}$, and $R_o = 1 \text{ }\Omega$ is driven by a source with $R_s = 25 \text{ k}\Omega$ and drives a load $R_L = 3 \text{ }\Omega$. Calculate the overall gain as well as the amount of input and output loading. (b) Repeat, but for a source with $R_s = 50 \text{ k}\Omega$ and a load $R_L = 4 \text{ }\Omega$. Compare.

Solution.

- (a) By Eq. (1.3), the overall gain is $v_O/v_S = [100/(25 + 100)] \times 100 \times 3/(1 + 3) = 0.80 \times 100 \times 0.75 = 60 \text{ V/V}$, which is less than 100 V/V because of loading. Input loading causes the source voltage to drop to 80% of its unloaded value; output loading introduces an additional drop to 75%.
- (b) By the same equation, $v_O/v_S = 0.67 \times 100 \times 0.80 = 53.3 \text{ V/V}$. We now have more loading at the input but less loading at the output. Moreover, the overall gain has changed from 60 V/V to 53.3 V/V .

Loading is generally undesirable because it makes the overall gain dependent on the particular input source and output load, not to mention gain reduction. The origin of loading is obvious: when the amplifier is connected to the input source, R_i draws current and causes R_s to drop some voltage. It is precisely this drop that, once subtracted from v_S , leads to a reduced voltage v_I . Likewise, at the output port the magnitude of v_O is less than the dependent-source voltage $A_{oc}v_I$ because of the voltage drop across R_o .

If loading could be eliminated altogether, we would have $v_O/v_S = A_{oc}$ regardless of the input source and the output load. To achieve this condition, the voltage drops across R_s and R_o must be zero regardless of R_s and R_L . The only way to achieve this is by requiring that our voltage amplifier have $R_i = \infty$ and $R_o = 0$. For obvious reasons such an amplifier is termed *ideal*. Though these conditions cannot be met in practice, an amplifier designer will strive to approximate them as closely as possible by ensuring that $R_i \gg R_s$ and $R_o \ll R_L$ for all input sources and output loads that the amplifier is likely to be connected to.

Another popular amplifier is the *current amplifier*. Since we are now dealing with currents, we model the input source and the amplifier with Norton equivalents, as in Fig. 1.2. The parameter A_{sc} of the current-controlled current source (CCCS)

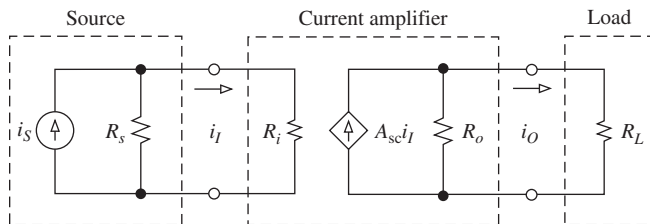


FIGURE 1.2
Current amplifier.

is called the *unloaded*, or *short-circuit*, *current gain*. Applying the current divider formula twice yields the source-to-load gain,

$$\frac{i_O}{i_S} = \frac{R_S}{R_S + R_i} A_{sc} \frac{R_o}{R_o + R_L} \quad (1.4)$$

We again witness loading both at the input port, where part of i_S is lost through R_S , making i_I less than i_S , and at the output port, where part of $A_{sc}i_I$ is lost through R_o . Consequently, we always have $|i_O/i_S| \leq |A_{sc}|$. To eliminate loading, an *ideal* current amplifier has $R_i = 0$ and $R_o = \infty$, exactly the opposite of the ideal voltage amplifier.

An amplifier whose input is a voltage v_I and whose output is a current i_O is called a *transconductance amplifier* because its gain is in amperes per volt, the dimensions of conductance. The situation at the input port is the same as that of the voltage amplifier of Fig. 1.1; the situation at the output port is similar to that of the current amplifier of Fig. 1.2, except that the dependent source is now a voltage-controlled current source (VCCS) of value $A_g v_I$, with A_g in amperes per volt. To avoid loading, an ideal transconductance amplifier has $R_i = \infty$ and $R_o = \infty$.

Finally, an amplifier whose input is a current i_I and whose output is a voltage v_O is called a *transresistance amplifier*, and its gain is in volts per ampere. The input port appears as in Fig. 1.2, and the output port as in Fig. 1.1, except that we now have a current-controlled voltage source (CCVS) of value $A_r i_I$, with A_r in volts per ampere. Ideally, such an amplifier has $R_i = 0$ and $R_o = 0$, the opposite of the transconductance amplifier.

The four basic amplifier types, along with their ideal input and output resistances, are summarized in Table 1.1.

TABLE 1.1
Basic amplifiers and their ideal terminal resistances

Input	Output	Amplifier type	Gain	R_i	R_o
v_I	v_O	Voltage	V/V	∞	0
i_I	i_O	Current	A/A	0	∞
v_I	i_O	Transconductance	A/V	∞	∞
i_I	v_O	Transresistance	V/A	0	0

1.2
THE OPERATIONAL AMPLIFIER

The operational amplifier is a voltage amplifier with extremely high gain. For example, the popular 741 op amp has a typical gain of 200,000 V/V, also expressed as 200 V/mV. Gain is also expressed in decibels (dB) as $20 \log_{10} 200,000 = 106$ dB. The OP77, a more recent type, has a gain of 12 million, or 12 V/ μ V, or $20 \log_{10}(12 \times 10^6) = 141.6$ dB. In fact, what distinguishes op amps from all other voltage amplifiers is the size of their gain. In the next sections we shall see that the higher the gain the better, or that an op amp would ideally have an infinitely large gain. Why one would want gain to be extremely large, let alone infinite, will become clearer as soon as we start analyzing our first op amp circuits.

Figure 1.3a shows the symbol of the op amp and the power-supply connections to make it work. The inputs, identified by the “−” and “+” symbols, are designated *inverting* and *noninverting*. Their voltages with respect to ground are denoted v_N and v_P , and the output voltage as v_O . The arrowhead signifies signal flow from the inputs to the output.

Op amps do not have a 0-V ground terminal. Ground reference is established externally by the power-supply common. The supply voltages are denoted V_{CC} and V_{EE} in the case of bipolar devices, and V_{DD} and V_{SS} in the case of CMOS devices. The typical dual-supply values of ± 15 V of the 741 days have been gradually reduced by over a decade, to the point that nowadays supplies of ± 1.25 V, or +1.25 V and 0 V, are not uncommon, especially in portable equipment. As we proceed, we shall use a variety of power-supply values, keeping in mind that most principles and applications you are about to learn are not critically dependent on the particular supplies in use. To minimize cluttering in circuit diagrams, it is customary not to show the power-supply connections. However, when we try out an op amp in the lab, we must remember to apply power to make it function.

Figure 1.3b shows the equivalent circuit of a properly powered op amp. Though the op amp itself does not have a ground pin, the ground symbol inside its equivalent circuit models the power-supply common of Fig. 1.3a. The equivalent circuit includes the *differential input resistance* r_d , the *voltage gain* a , and the *output resistance* r_o . For reasons that will become clear in the next sections, r_d , a , and r_o are referred to

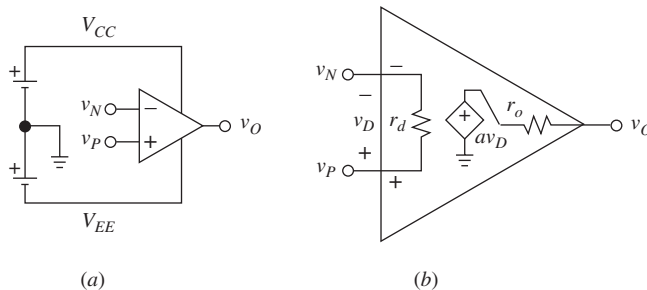


FIGURE 1.3

(a) Op amp symbol and power-supply connections. (b) Equivalent circuit of a powered op amp. (The 741 op amp has typically $r_d = 2 \text{ M}\Omega$, $a = 200 \text{ V/mV}$, and $r_o = 75 \Omega$.)

as *open-loop* parameters and are symbolized by lowercase letters. The difference

$$v_D = v_P - v_N \quad (1.5)$$

is called the *differential input voltage*, and gain a is also called the *unloaded gain* because in the absence of output loading we have

$$v_O = av_D = a(v_P - v_N) \quad (1.6)$$

Since both input terminals are allowed to attain independent potentials with respect to ground, the input port is said to be of the *double-ended* type. Contrast this with the output port, which is of the *single-ended* type. Equation (1.6) indicates that the op amp responds only to the difference between its input voltages, not to their individual values. Consequently, op amps are also called *difference amplifiers*.

Reversing Eq. (1.6), we obtain

$$v_D = \frac{v_O}{a} \quad (1.7)$$

which allows us to find the voltage v_D causing a given v_O . We again observe that this equation yields only the difference v_D , not the values of v_N and v_P themselves. Because of the large gain a in the denominator, v_D is bound to be very small. For instance, to sustain $v_O = 6$ V, an unloaded 741 op amp needs $v_D = 6/200,000 = 30 \mu\text{V}$, quite a small voltage. An unloaded OP77 would need $v_D = 6/(12 \times 10^6) = 0.5 \mu\text{V}$, an even smaller value!

The Ideal Op Amp

We know that to minimize loading, a well-designed voltage amplifier must draw negligible (ideally zero) current from the input source and must present negligible (ideally zero) resistance to the output load. Op amps are no exception, so we define the ideal op amp as an ideal voltage amplifier with infinite open-loop gain:

$$a \rightarrow \infty \quad (1.8a)$$

Its ideal terminal conditions are

$$r_d = \infty \quad (1.8b)$$

$$r_o = 0 \quad (1.8c)$$

$$i_P = i_N = 0 \quad (1.8d)$$

where i_P and i_N are the currents drawn by the noninverting and inverting inputs. The ideal op amp model is shown in Fig. 1.4.

We observe that in the limit $a \rightarrow \infty$, we obtain $v_D \rightarrow v_O/\infty \rightarrow 0$! This result is often a source of puzzlement because it makes one wonder how an amplifier with zero input can sustain a nonzero output. Shouldn't the output also be zero by Eq. (1.6)? The answer lies in the fact that as gain a approaches infinity, v_D does indeed approach zero, but in such a way as to maintain the product av_D nonzero and equal to v_O .

Real-life op amps depart somewhat from the ideal, so the model of Fig. 1.4 is only a conceptualization. But during our initiation into the realm of op amp circuits, we shall use this model because it relieves us from worrying about loading effects

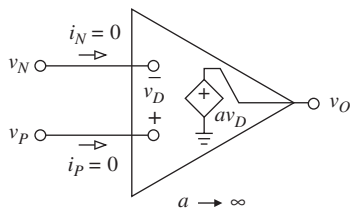


FIGURE 1.4
Ideal op amp model.

so that we can concentrate on the role of the op amp itself. Once we have developed enough understanding and confidence, we shall backtrack and use the more realistic model of Fig. 1.3b to assess the validity of our results. We shall find that the results obtained with the ideal and with the real-life models are in much closer agreement than we might have suspected, corroborating the claim that the ideal model, though a conceptualization, is not that academic after all.

SPICE Simulation

Circuit simulation by computer has become a powerful and indispensable tool in both analysis and design. In this book we shall use SPICE, both to verify our calculations and to investigate higher-order effects that would be too complex for paper-and-pencil analysis. The reader is assumed to be conversant with the SPICE basics covered in prerequisite courses. SPICE is available in a wide variety of versions under continuous revision. Though the circuit examples of this book were created using the Student Version of Cadence's PSpice, the reader can easily redraw and rerun them in the version of SPICE in his/her possession.

We begin with the basic model of Fig. 1.5, which reflects 741 data. The circuit uses a voltage-controlled voltage source (VCVS) to model voltage gain, and a resistor pair to model the terminal resistances (by PSpice convention, the “+” input is shown at the top and the “−” input at the bottom, just the opposite of op amp convention).

If a pseudo-ideal model is desired, then r_d is left open, r_o is shorted out, and the source value is increased from 200 kV/V to some huge value, say, 1 GV/V. (However, the reader is cautioned that too large a value may cause convergence problems.)

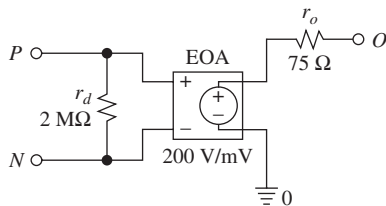


FIGURE 1.5
Basic SPICE model of the 741 op amp.

By connecting external components around an op amp, we obtain what we shall henceforth refer to as an *op amp circuit*. It is crucial that you understand the difference between an op amp circuit and a plain op amp. Think of the latter as a component of the former, just as the external components are. The most basic op amp circuits are the *inverting*, *noninverting*, and *buffer amplifiers*.

The Noninverting Amplifier

The circuit of Fig. 1.6a consists of an op amp and two external resistors. To understand its function, we need to find a relationship between v_O and v_I . To this end we redraw it as in Fig. 1.6b, where the op amp has been replaced by its equivalent model and the resistive network has been rearranged to emphasize its role in the circuit. We can find v_O via Eq. (1.6); however, we must first derive expressions for v_P and v_N . By inspection,

$$v_P = v_I \quad (1.9)$$

Using the voltage divider formula yields $v_N = [R_1/(R_1 + R_2)]v_O$, or

$$v_N = \frac{1}{1 + R_2/R_1} v_O \quad (1.10)$$

The voltage v_N represents the fraction of v_O that is being fed back to the inverting input. Consequently, the function of the resistive network is to create *negative feedback* around the op amp. Letting $v_O = a(v_P - v_N)$, we get

$$v_O = a \left(v_I - \frac{1}{1 + R_2/R_1} v_O \right) \quad (1.11)$$

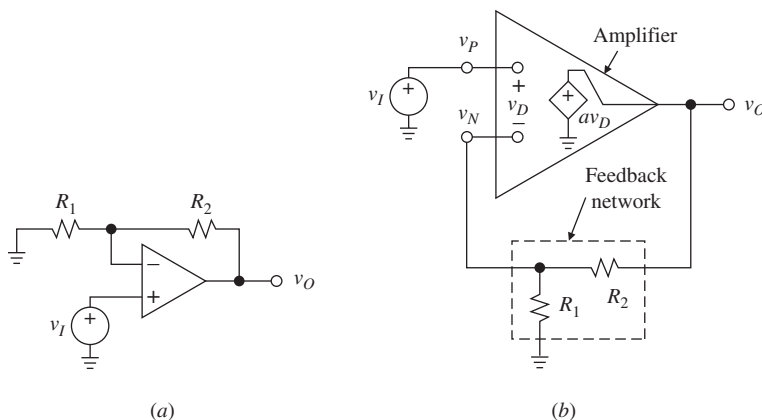


FIGURE 1.6
Noninverting amplifier and circuit model for its analysis.

Collecting terms and solving for the ratio v_O/v_I , which we shall designate as A , yields, after minor rearrangement,

$$A = \frac{v_O}{v_I} = \left(1 + \frac{R_2}{R_1}\right) \frac{1}{1 + (1 + R_2/R_1)/a} \quad (1.12)$$

This result reveals that the circuit of Fig. 1.6a, consisting of an op amp plus a resistor pair, is itself an amplifier, and that its gain is A . Since A is positive, the polarity of v_O is the same as that of v_I —hence the name *noninverting amplifier*.

The gain A of the op amp circuit and the gain a of the basic op amp are quite different. This is not surprising, as the two amplifiers, while sharing the same output v_O , have different inputs, namely, v_I for the former and v_D for the latter. To underscore this difference, a is referred to as the *open-loop gain*, and A as the *closed-loop gain*, the latter designation stemming from the fact that the op amp circuit contains a loop. In fact, starting from the inverting input in Fig. 1.6b, we can trace a clockwise loop through the op amp and then through the resistive network, which brings us back to the starting point.

EXAMPLE 1.2. In the circuit of Fig. 1.6a, let $v_I = 1$ V, $R_1 = 2$ k Ω , and $R_2 = 18$ k Ω . Find v_O if (a) $a = 10^2$ V/V, (b) $a = 10^4$ V/V, (c) $a = 10^6$ V/V. Comment on your findings.

Solution. Equation (1.12) gives $v_O/1 = (1 + 18/2)/(1 + 10/a)$, or $v_O = 10/(1 + 10/a)$. So

(a) $v_O = 10/(1 + 10/10^2) = 9.091$ V,

(b) $v_O = 9.990$ V,

(c) $v_O = 9.9999$ V.

The higher the gain a , the closer v_O is to 10.0 V.

Ideal Noninverting Amplifier Characteristics

Letting $a \rightarrow \infty$ in Eq. (1.12) yields a closed-loop gain that we refer to as ideal:

$$A_{\text{ideal}} = \lim_{a \rightarrow \infty} A = 1 + \frac{R_2}{R_1} \quad (1.13)$$

In this limit A becomes independent of a , and its value is set exclusively by the *external resistance ratio*, R_2/R_1 . We can now appreciate the reason for wanting $a \rightarrow \infty$. Indeed, a circuit whose closed-loop gain depends only on a resistance ratio offers tremendous advantages for the designer since it makes it easy to tailor gain to the application at hand. For instance, suppose you need an amplifier with a gain of 2 V/V. Then, by Eq. (1.13), pick $R_2/R_1 = A - 1 = 2 - 1 = 1$; for example, pick $R_1 = R_2 = 100$ k Ω . Do you want $A = 10$ V/V? Then pick $R_2/R_1 = 9$; for example, $R_1 = 20$ k Ω and $R_2 = 180$ k Ω . Do you want an amplifier with variable gain? Then make R_1 or R_2 variable by means of a potentiometer (pot). For example, if R_1 is a fixed 10-k Ω resistor and R_2 is a 100-k Ω pot configured as a variable resistance from 0 Ω to 100 k Ω , then Eq. (1.13) indicates that the gain can be varied over the range $1 \text{ V/V} \leq A \leq 11 \text{ V/V}$. No wonder it is desirable that $a \rightarrow \infty$. It leads to the simpler expression of Eq. (1.13), and it makes op amp circuit design a real snap!

Another advantage of Eq. (1.13) is that gain A can be made as accurate and stable as needed by using resistors of suitable quality. Actually it is not even necessary that the individual resistors be of high quality; it only suffices that their ratio be so. For example, using two resistances that track each other with temperature so as to maintain a constant ratio will make gain A temperature-independent. Contrast this with gain a , which depends on the characteristics of the resistors, diodes, and transistors inside the op amp, and is therefore sensitive to thermal drift, aging, and production variations. This is a prime example of one of the most fascinating aspects of electronics, namely, the ability to implement high-performance circuits using inferior components!

The advantages afforded by Eq. (1.13) do not come for free. The price is the size of gain a needed to make this equation acceptable within a given degree of accuracy (more on this will follow). It is often said that we are in effect throwing away a good deal of open-loop gain for the sake of stabilizing the closed-loop gain. Considering the benefits, the price is well worth paying, especially with IC technology, which, in mass production, makes it possible to achieve high open-loop gains at extremely low cost.

Since the op amp circuit of Fig. 1.6 has proven to be an amplifier itself, besides gain A it must also present input and output resistances, which we shall designate as R_i and R_o and call the *closed-loop input* and *output resistances*. You may have noticed that to keep the distinction between the parameters of the basic op amp and those of the op amp circuit, we are using lowercase letters for the former and uppercase letters for the latter.

Though we shall have more to say about R_i and R_o from the viewpoint of negative feedback in Section 1.6, we presently use the simplified model of Fig. 1.6b to state that $R_i = \infty$ because the noninverting input terminal appears as an open circuit, and $R_o = 0$ because the output comes directly from the source av_D . In summary,

$$R_i = \infty \quad R_o = 0 \quad (1.14)$$

which, according to Table 1.1, represent the ideal terminal characteristics of a voltage amplifier. The equivalent circuit of the ideal noninverting amplifier is shown in Fig. 1.7.

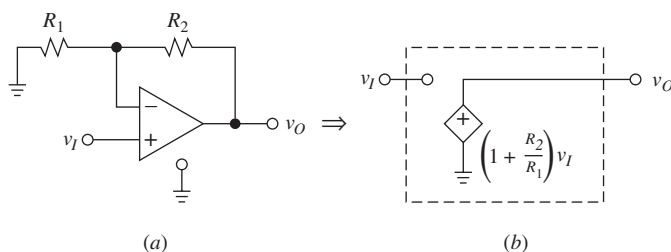


FIGURE 1.7
Noninverting amplifier and its ideal equivalent circuit.

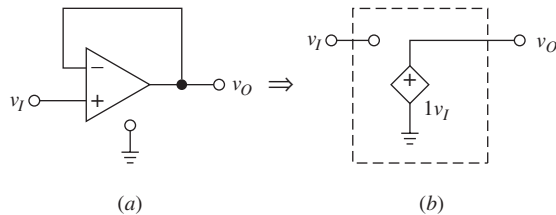


FIGURE 1.8
Voltage follower and its ideal equivalent circuit.

The Voltage Follower

Letting $R_1 = \infty$ and $R_2 = 0$ in the noninverting amplifier turns it into the *unity-gain amplifier*, or *voltage follower*, of Fig. 1.8a. Note that the circuit consists of the op amp and a wire to feed the entire output back to the input. The closed-loop parameters are

$$A = 1 \text{ V/V} \quad R_i = \infty \quad R_o = 0 \quad (1.15)$$

and the equivalent circuit is shown in Fig. 1.8b. As a voltage amplifier, the follower is not much of an achiever since its gain is only unity. Its specialty, however, is to act as a *resistance transformer*, since looking into its input we see an open circuit, but looking into its output we see a short circuit to a source of value $v_O = v_I$.

To appreciate this feature, consider a source v_S whose voltage we wish to apply across a load R_L . If the source were ideal, all we would need would be a plain wire to connect the two. However, if the source has nonzero output resistance R_s , as in Fig. 1.9a, then R_s and R_L will form a voltage divider and the magnitude of v_L will be less than that of v_S because of the voltage drop across R_s . Let us now replace the wire by a voltage follower as in Fig. 1.9b. Since the follower has $R_i = \infty$, there is no loading at the input, so $v_I = v_S$. Moreover, since the follower has $R_o = 0$, loading is absent also from the output, so $v_L = v_I = v_S$, indicating that R_L now receives the full source voltage with no losses. The role of the follower is thus to act as a *buffer* between source and load.

We also observe that now the source delivers no current and hence no power, while in the circuit of Fig. 1.9a, it did. The current and power drawn by R_L are now supplied by the op amp, which in turn takes them from its power supplies,

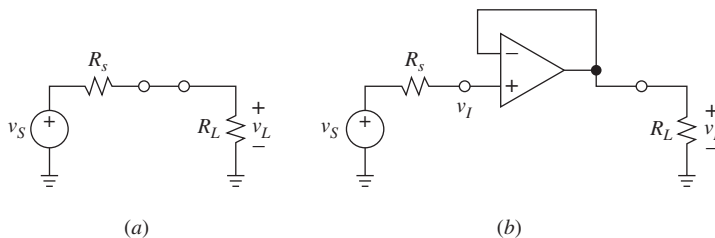


FIGURE 1.9
Source and load connected (a) directly, and (b) via a voltage follower to eliminate loading.

not explicitly shown in the figure. Thus, besides restoring v_L to the full value of v_S , the follower relieves the source v_S from supplying any power. The need for a buffer arises so often in electronic design that special circuits are available whose performance has been optimized for this function. The BUF03 is a popular example.

The Inverting Amplifier

Together with the noninverting amplifier, the inverting configuration of Fig. 1.10a constitutes a cornerstone of op amp applications. The inverting amplifier was invented before the noninverting amplifier because in their early days op amps had only one input, namely, the inverting one. Referring to the equivalent circuit of Fig. 1.10b, we have

$$v_P = 0 \quad (1.16)$$

Applying the superposition principle yields $v_N = [R_2/(R_1 + R_2)]v_I + [R_1/(R_1 + R_2)]v_O$, or

$$v_N = \frac{1}{1 + R_1/R_2}v_I + \frac{1}{1 + R_2/R_1}v_O \quad (1.17)$$

Letting $v_O = a(v_P - v_N)$ yields

$$v_O = a \left(-\frac{1}{1 + R_1/R_2}v_I - \frac{1}{1 + R_2/R_1}v_O \right) \quad (1.18)$$

Comparing with Eq. (1.11), we observe that the resistive network still feeds the portion $1/(1 + R_2/R_1)$ of v_O back to the inverting input, thus providing the same amount of negative feedback. Solving for the ratio v_O/v_I and rearranging, we obtain

$$A = \frac{v_O}{v_I} = \left(-\frac{R_2}{R_1} \right) \frac{1}{1 + (1 + R_2/R_1)/a} \quad (1.19)$$

Our circuit is again an *amplifier*. However, the gain A is now *negative*, indicating that the polarity of v_O will be opposite to that of v_I . This is not surprising, because we are now applying v_I to the inverting side of the op amp. Hence, the circuit is called an *inverting amplifier*. If the input is a sine wave, the circuit will introduce a *phase reversal*, or, equivalently, a 180° *phase shift*.

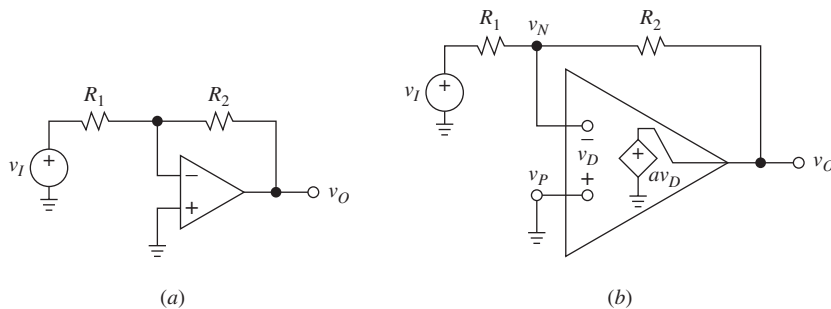


FIGURE 1.10
Inverting amplifier and circuit model for its analysis.

Ideal Inverting Amplifier Characteristics

Letting $a \rightarrow \infty$ in Eq. (1.19), we obtain

$$A_{\text{ideal}} = \lim_{a \rightarrow \infty} A = -\frac{R_2}{R_1} \quad (1.20)$$

That is, the closed-loop gain again depends only on an external resistance ratio, yielding well-known advantages for the circuit designer. For instance, if we need an amplifier with a gain of -5 V/V, we pick two resistances in a 5:1 ratio, such as $R_1 = 20$ k Ω and $R_2 = 100$ k Ω . If, on the other hand, R_1 is a fixed 20-k Ω resistor and R_2 is a 100-k Ω pot configured as a variable resistance, then the closed-loop gain can be varied anywhere over the range -5 V/V $\leq A \leq 0$. Note in particular that the magnitude of A can now be controlled all the way down to zero.

We now turn to the task of determining the closed-loop input and output resistances R_i and R_o . Since $v_D = v_O/a$ is vanishingly small because of the large size of a , it follows that v_N is very close to v_P , which is zero. In fact, in the limit $a \rightarrow \infty$, v_N would be zero exactly, and would be referred to as *virtual ground* because to an outside observer, things appear as if the inverting input were permanently grounded. We conclude that the effective resistance seen by the input source is just R_1 . Moreover, since the output comes directly from the source av_D , we have $R_o = 0$. In summary,

$$R_i = R_1 \quad R_o = 0 \quad (1.21)$$

The equivalent circuit of the inverting amplifier is shown in Fig. 1.11.

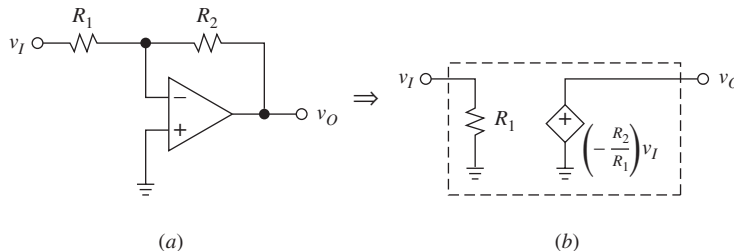


FIGURE 1.11
Inverting amplifier and its ideal equivalent circuit.

EXAMPLE 1.3. (a) Using the basic 741 model of Fig. 1.4, direct PSpice to find the closed-loop parameters of an inverting amplifier implemented with $R_1 = 1.0$ k Ω and $R_2 = 100$ k Ω . Compare with the ideal case and comment. (b) What happens if you raise a to 1 G V/V? Lower a to 1 kV/V?

Solution.

(a) After creating the circuit of Fig. 1.12, we direct PSpice to calculate the small-signal gain (**TF**) from the input source **V(I)** to the output variable **V(O)**. This yields the following output file:

```
V(O)/V(I) = -9.995E+01
INPUT RESISTANCE AT V(I) = 1.001E+03
OUTPUT RESISTANCE AT V(O) = 3.787E-02
```

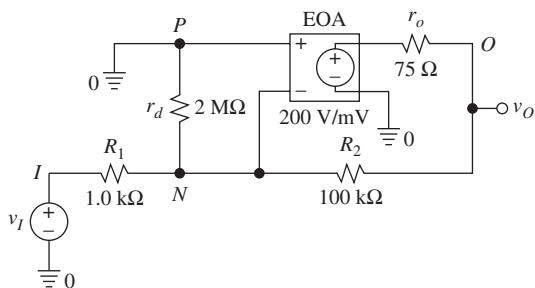


FIGURE 1.12
SPICE circuit for Example 1.3.

It is apparent that the data are quite close to the ideal values $A = -100$ V/V, $R_i = 1.0$ k Ω , and $R_o \rightarrow 0$.

- (b) Rerunning PSpice after increasing the dependent-source gain from 200 kV/V to 1 G V/V we find that A and R_i match the ideal values (within PSpice's resolution) and R_o drops to micro-ohms, which is very close to 0. Lowering the gain to 1 kV/V gives $A = -90.82$ V/V, $R_i = 1.100$ k Ω , and $R_o = 6.883$ Ω , indicating a more pronounced departure from the ideal.

Unlike its noninverting counterpart, the inverting amplifier will load down the input source if the source is nonideal. This is depicted in Fig. 1.13. Since in the limit $a \rightarrow \infty$, the op amp keeps $v_N \rightarrow 0$ V (virtual ground), we can apply the voltage divider formula and write

$$v_I = \frac{R_1}{R_s + R_1} v_S \quad (1.22)$$

indicating that $|v_I| \leq |v_S|$. Applying Eq. (1.20), $v_L/v_I = -R_2/R_1$. Eliminating v_I , we obtain

$$\frac{v_L}{v_S} = -\frac{R_2}{R_s + R_1} \quad (1.23)$$

Because of loading at the input, the magnitude of the overall gain, $R_2/(R_s + R_1)$, is less than that of the amplifier alone, R_2/R_1 . The amount of loading depends on the relative magnitudes of R_s and R_1 , and only if $R_s \ll R_1$ can loading be ignored.

We can look at the above circuit also from another viewpoint. Namely, to find the gain v_L/v_S , we can still apply Eq. (1.20), provided, however, that we regard R_s

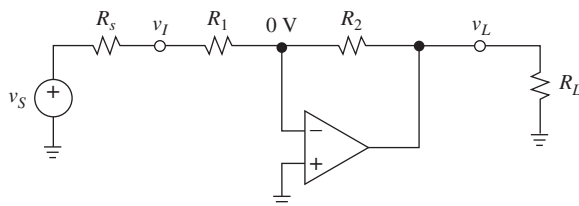


FIGURE 1.13
Input loading by the inverting amplifier.

and R_1 as a *single* resistance of value $R_s + R_1$. Thus, $v_L/v_S = -R_2/(R_s + R_1)$, the same as above.

1.4 IDEAL OP AMP CIRCUIT ANALYSIS

Considering the simplicity of the ideal closed-loop results of the previous section, we wonder whether there is not a simpler technique to derive them, bypassing some of the tedious algebra. Such a technique exists and is based on the fact that when the op amp is operated with negative feedback, in the limit $a \rightarrow \infty$ its input voltage $v_D = v_O/a$ approaches zero,

$$\lim_{a \rightarrow \infty} v_D = 0 \quad (1.24)$$

or, since $v_N = v_P - v_D = v_P - v_O/a$, v_N approaches v_P ,

$$\lim_{a \rightarrow \infty} v_N = v_P \quad (1.25)$$

This property, referred to as the *input voltage constraint*, makes the input terminals appear as if they were shorted together, though in fact they are not. We also know that an ideal op amp draws no current at its input terminals, so this apparent short carries no current, a property referred to as the *input current constraint*. In other words, for voltage purposes the input port appears as a short circuit, but for current purposes it appears as an open circuit! Hence the designation *virtual short*. Summarizing, *when operated with negative feedback, an ideal op amp will output whatever voltage and current it takes to drive v_D to zero or, equivalently, to force v_N to track v_P , but without drawing any current at either input terminal.*

Note that it is v_N that tracks v_P , not the other way around. The op amp controls v_N via the external feedback network. Without feedback, the op amp would be unable to influence v_N and the above equations would no longer hold.

To better understand the action of the op amp, consider the simple circuit of Fig. 1.14a, where we have, by inspection, $i = 0$, $v_1 = 0$, $v_2 = 6$ V, and $v_3 = 6$ V. If we now connect an op amp as in Fig. 1.14b, what will happen? As we know, the op amp will drive v_3 to whatever it takes to make $v_2 = v_1$. To find these voltages,

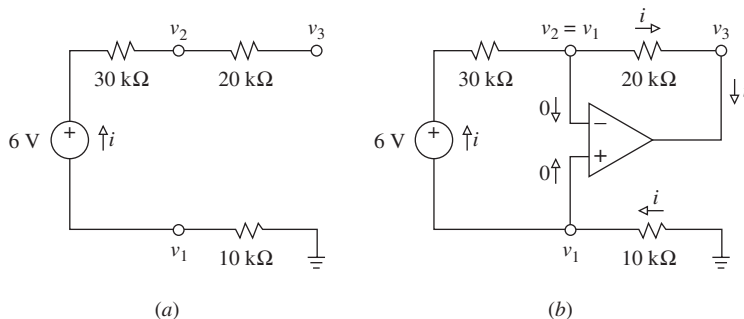


FIGURE 1.14
The effect of an op amp in a circuit.

we equate the current entering the 6-V source to that exiting it; or

$$\frac{0 - v_1}{10} = \frac{(v_1 + 6) - v_2}{30}$$

Letting $v_2 = v_1$ and solving yields $v_1 = -2$ V. The current is

$$i = \frac{0 - v_1}{10} = \frac{2}{10} = 0.2 \text{ mA}$$

and the output voltage is

$$v_3 = v_2 - 20i = -2 - 20 \times 0.2 = -6 \text{ V}$$

Summarizing, as the op amp is inserted in the circuit, it swings v_3 from 6 V to -6 V because this is the voltage that makes $v_2 = v_1$. Consequently, v_1 is changed from 0 V to -2 V, and v_2 from 6 V to -2 V. The op amp also sinks a current of 0.2 mA at its output terminal, but without drawing any current at either input.

The Basic Amplifiers Revisited

It is instructive to derive the noninverting and inverting amplifier gains using the concept of the *virtual short*. In the circuit of Fig. 1.15a we exploit this concept to label the inverting-input voltage as v_I . Applying the voltage divider formula, we have $v_I = v_O / (1 + R_2/R_1)$, which is readily turned around to yield the familiar relationship $v_O = (1 + R_2/R_1)v_I$. In words, the noninverting amplifier provides the *inverse* function of the voltage divider: the divider attenuates v_O to yield v_I , whereas the amplifier magnifies v_I by the inverse amount to yield v_O . This action can be visualized via the lever analog depicted above the amplifier in the figure. The lever pivots around a point corresponding to ground. The lever segments correspond to resistances, and the swings correspond to voltages.

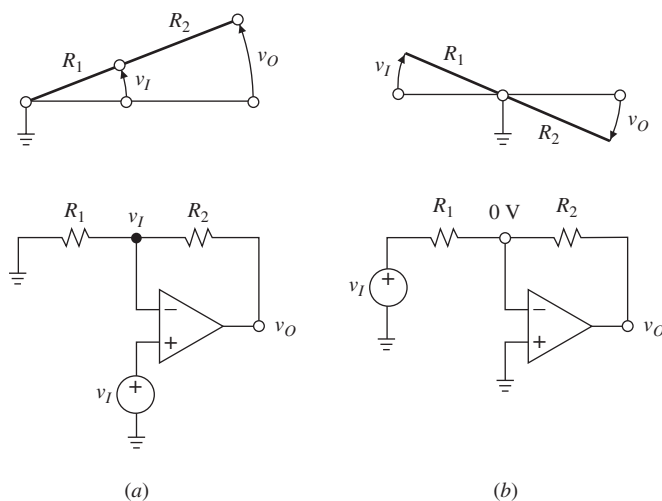


FIGURE 1.15
Mechanical analogies of the noninverting and the inverting amplifiers.

In the circuit of Fig. 1.15*b* we again exploit the virtual-short concept to label the inverting input as a virtual ground, or 0 V. Applying Kirchhoff's current law (KCL), we have $(v_I - 0)/R_1 = (0 - v_O)/R_2$, which is readily solved for v_O to yield the familiar relationship $v_O = (-R_2/R_1)v_I$. This can be visualized via the mechanical analog shown above the amplifier. An upswing (downswing) at the input produces a downswing (upswing) at the output. By contrast, in Fig. 1.15*a* the output swings in the same direction as the input.

So far, we have only studied the basic op amp configurations. It is time to familiarize ourselves with other op amp circuits. These we shall study using the virtual-short concept.

The Summing Amplifier

The summing amplifier has two or more inputs and one output. Though the example of Fig. 1.16 has three inputs, v_1 , v_2 , and v_3 , the following analysis can readily be generalized to an arbitrary number of them. To obtain a relationship between output and inputs, we impose that the total current entering the virtual-ground node equal that exiting it, or

$$i_1 + i_2 + i_3 = i_F$$

For obvious reasons, this node is also referred to as a *summing junction*. Using Ohm's law, $(v_1 - 0)/R_1 + (v_2 - 0)/R_2 + (v_3 - 0)/R_3 = (0 - v_O)/R_F$, or

$$\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = -\frac{v_O}{R_F}$$

We observe that thanks to the virtual ground, the input currents are linearly proportional to the corresponding source voltages. Moreover, the sources are prevented from interacting with each other—a very desirable feature should any of these sources be

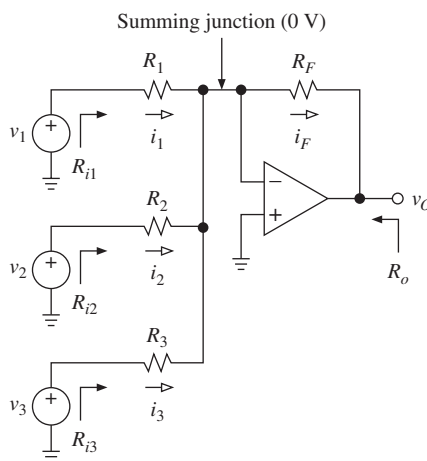


FIGURE 1.16
 Summing amplifier.

disconnected from the circuit. Solving for v_O yields

$$v_O = - \left(\frac{R_F}{R_1} v_1 + \frac{R_F}{R_2} v_2 + \frac{R_F}{R_3} v_3 \right) \quad (1.26)$$

indicating that the output is a weighted sum of the inputs (hence the name *summing amplifier*), with the weights being established by resistance ratios. A popular application of summing amplifiers is audio mixing.

Since the output comes directly from the dependent source inside the op amp, we have $R_o = 0$. Moreover, because of the virtual ground, the input resistance R_{ik} ($k = 1, 2, 3$) seen by source v_k equals the corresponding resistance R_k . In summary,

$$\begin{aligned} R_{ik} &= R_k & k &= 1, 2, 3 \\ R_o &= 0 \end{aligned} \quad (1.27)$$

If the input sources are nonideal, the circuit will load them down, as in the case of the inverting amplifier. Equation (1.26) is still applicable provided we replace R_k by $R_{sk} + R_k$ in the denominators, where R_{sk} is the output resistance of the k th input source.

EXAMPLE 1.4. Using standard 5% resistances, design a circuit such that $v_O = -2(3v_1 + 4v_2 + 2v_3)$.

Solution. By Eq. (1.26) we have $R_F/R_1 = 6$, $R_F/R_2 = 8$, $R_F/R_3 = 4$. One possible standard resistance set satisfying the above conditions is $R_1 = 20 \text{ k}\Omega$, $R_2 = 15 \text{ k}\Omega$, $R_3 = 30 \text{ k}\Omega$, and $R_F = 120 \text{ k}\Omega$.

EXAMPLE 1.5. In the design of function generators and data converters, the need arises to *offset* as well as *amplify* a given voltage v_I to obtain a voltage of the type $v_O = Av_I + V_O$, where V_O is the desired amount of offset. An offsetting amplifier can be implemented with a summing amplifier in which one of the inputs is v_I and the other is either V_{CC} or V_{EE} , the regulated supply voltages used to power the op amp. Using standard 5% resistances, design a circuit such that $v_O = -10v_I + 2.5 \text{ V}$. Assume $\pm 5\text{-V}$ supplies.

Solution. The circuit is shown in Fig. 1.17. Imposing $v_O = -(R_F/R_1)v_I - (R_F/R_2) \times (-5) = -10v_I + 2.5$, we find that a possible resistance set is $R_1 = 10 \text{ k}\Omega$, $R_2 = 200 \text{ k}\Omega$, and $R_F = 100 \text{ k}\Omega$, as shown.

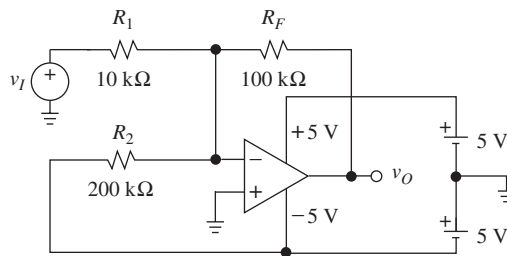


FIGURE 1.17
A dc-offsetting amplifier.

If $R_3 = R_2 = R_1$, then Eq. (1.26) yields

$$v_O = -\frac{R_F}{R_1}(v_1 + v_2 + v_3) \quad (1.28)$$

that is, v_O is proportional to the *true sum* of the inputs. The proportionality constant $-R_F/R_1$ can be varied all the way down to zero by implementing R_F with a variable resistance. If all resistances are equal, the circuit yields the (inverted) sum of its inputs, $v_O = -(v_1 + v_2 + v_3)$.

The Difference Amplifier

As shown in Fig. 1.18, the difference amplifier has one output and two inputs, one of which is applied to the inverting side, the other to the noninverting side. We can find v_O via the superposition principle as $v_O = v_{O1} + v_{O2}$, where v_{O1} is the value of v_O with v_2 set to zero, and v_{O2} that with v_1 set to zero.

Letting $v_2 = 0$ yields $v_P = 0$, making the circuit act as an inverting amplifier with respect to v_1 . So $v_{O1} = -(R_2/R_1)v_1$ and $R_{i1} = R_1$, where R_{i1} is the input resistance seen by the source v_1 .

Letting $v_1 = 0$ makes the circuit act as a noninverting amplifier with respect to v_P . So $v_{O2} = (1 + R_2/R_1)v_P = (1 + R_2/R_1) \times [R_4/(R_3 + R_4)]v_2$ and $R_{i2} = R_3 + R_4$, where R_{i2} is the input resistance seen by the source v_2 . Letting $v_O = v_{O1} + v_{O2}$ and rearranging yields

$$v_O = \frac{R_2}{R_1} \left(\frac{1 + R_1/R_2}{1 + R_3/R_4} v_2 - v_1 \right) \quad (1.29)$$

Moreover,

$$R_{i1} = R_1 \quad R_{i2} = R_3 + R_4 \quad R_o = 0 \quad (1.30)$$

The output is again a linear combination of the inputs, but with coefficients of opposite polarity because one input is applied to the inverting side and the other to the noninverting side of the op amp. Moreover, the resistances seen by the input sources are finite and, in general, different from each other. If these sources are nonideal, the circuit will load them down, generally by different amounts. Let the

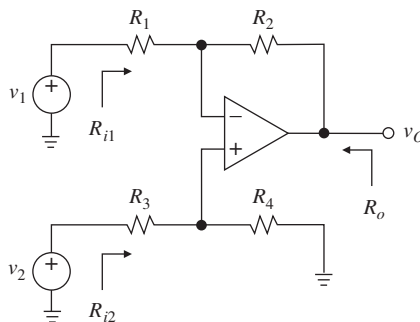


FIGURE 1.18
Difference amplifier.

sources have output resistances R_{s1} and R_{s2} . Then Eq. (1.29) is still applicable provided we replace R_1 by $R_{s1} + R_1$ and R_3 by $R_{s2} + R_3$.

EXAMPLE 1.6. Design a circuit such that $v_O = v_2 - 3v_1$ and $R_{i1} = R_{i2} = 100 \text{ k}\Omega$.

Solution. By Eq. (1.30) we must have $R_1 = R_{i1} = 100 \text{ k}\Omega$. By Eq. (1.29) we must have $R_2/R_1 = 3$, so $R_2 = 300 \text{ k}\Omega$. By Eq. (1.30) $R_3 + R_4 = R_{i2} = 100 \text{ k}\Omega$. By Eq. (1.29), $3[(1 + 1/3)/(1 + R_3/R_4)] = 1$. Solving the last two equations for their two unknowns yields $R_3 = 75 \text{ k}\Omega$ and $R_4 = 25 \text{ k}\Omega$.

An interesting case arises when the resistance pairs in Fig. 1.18 are in equal ratios:

$$\frac{R_3}{R_4} = \frac{R_1}{R_2} \quad (1.31)$$

When this condition is met, the resistances are said to form a *balanced bridge*, and Eq. (1.29) simplifies to

$$v_O = \frac{R_2}{R_1}(v_2 - v_1) \quad (1.32)$$

The output is now proportional to the *true difference* of the inputs—hence the name of the circuit. A popular application of the true difference amplifier is as a building block of instrumentation amplifiers, to be studied in the next chapter.

The Differentiator

To find the input-output relationship for the circuit of Fig. 1.19, we start out by imposing $i_C = i_R$. Using the capacitance law and Ohm's law, this becomes $Cd(v_I - 0)/dt = (0 - v_O)/R$, or

$$v_O(t) = -RC \frac{dv_I(t)}{dt} \quad (1.33)$$

The circuit yields an output that is proportional to the *time derivative* of the input—hence the name. The proportionality constant is set by R and C , and its units are seconds (s).

If you try out the differentiator circuit in the lab, you will find that it tends to oscillate. Its stability problems stem from the open-loop gain rolloff with frequency, an issue that will be addressed in Chapter 8. Suffice it to say here that the circuit is usually stabilized by placing a suitable resistance R_s in series with C . After this

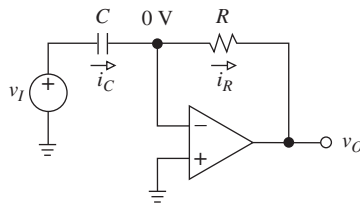


FIGURE 1.19
The op amp differentiator.

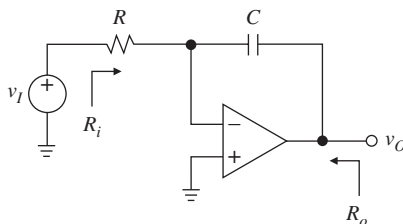


FIGURE 1.20
The op amp integrator.

modification the circuit will still provide the differentiation function, but only over a limited frequency range.

The Integrator

The analysis of the circuit of Fig. 1.20 mirrors that of Fig. 1.19. Imposing $i_R = i_C$, we now get $(v_I - 0)/R = C d(0 - v_O)/dt$, or $dv_O(t) = (-1/RC)v_I(t) dt$. Changing t to the dummy integration variable ξ and then integrating both sides from 0 to t yields

$$v_O(t) = -\frac{1}{RC} \int_0^t v_I(\xi) d\xi + v_O(0) \quad (1.34)$$

where $v_O(0)$ is the value of the output at $t = 0$. This value depends on the charge initially stored in the capacitor. Equation (1.34) indicates that the output is proportional to the *time integral* of the input—hence the name. The proportionality constant is set by R and C , but its units are now s^{-1} . Mirroring the analysis of the inverting amplifier, you can readily verify that

$$R_i = R \quad R_o = 0 \quad (1.35)$$

Thus, if the driving source has an output resistance R_s , in order to apply Eq. (1.34) we must replace R with $R_s + R$.

The op amp integrator, also called a *precision integrator* because of the high degree of accuracy with which it can implement Eq. (1.34), is a workhorse of electronics. It finds wide application in function generators (triangle and sawtooth wave generators), active filters (state-variable and biquad filters, switched-capacitor filters), analog-to-digital converters (dual-slope converters, quantized-feedback converters), and analog controllers (PID controllers).

If $v_I(t) = 0$, Eq. (1.34) predicts that $v_O(t) = v_O(0) = \text{constant}$. In practice, when the integrator circuit is tried out in the lab, it is found that its output will drift until it saturates at a value close to one of the supply voltages, even with v_I grounded. This is due to the so-called input offset error of the op amp, an issue to be discussed in Chapter 5. Suffice it to say here that a crude method of preventing saturation is to place a suitable resistance R_p in parallel with C . The resulting circuit, called a *lossy integrator*, will still provide the integration function, but only over a limited frequency range. Fortunately, in most applications integrators are placed inside a control loop designed to automatically keep the circuit away from saturation, at least

under proper operating conditions, thus eliminating the need for the aforementioned parallel resistance.

The Negative-Resistance Converter (NIC)

We conclude by demonstrating another important op amp application besides signal processing, namely, *impedance transformation*. To illustrate, consider the plain resistance of Fig. 1.21a. To find its value experimentally, we apply a test source v , we measure the current i out of the source's positive terminal, and then we let $R_{eq} = v/i$, where R_{eq} is the value of the resistance as seen by the source. Clearly, in this simple case $R_{eq} = R$. Moreover, the test source releases power and the resistance absorbs power.

Suppose we now lift the lower terminal of R off ground and drive it with a noninverting amplifier with the input tied to the other terminal of R , as shown in Fig. 1.21b. The current is now $i = [v - (1 + R_2/R_1)v]/R = -R_2v/(R_1R)$. Letting $R_{eq} = v/i$ yields

$$R_{eq} = -\frac{R_1}{R_2}R \quad (1.36)$$

indicating that the circuit simulates a *negative resistance*. The meaning of the negative sign is that current is now actually flowing *into* the test source's positive terminal, causing the source to absorb power. Consequently, a negative resistance *releases* power.

If $R_1 = R_2$, then $R_{eq} = -R$. In this case the test voltage v is amplified to $2v$ by the op amp, making R experience a net voltage v , positive at the right. Consequently, $i = -v/R = v/(-R)$.

Negative resistances can be used to neutralize unwanted ordinary resistances, as in the design of current sources, or to control pole location, as in the design of active filters and oscillators.

Looking back at the circuits covered so far, note that by interconnecting suitable components around a high-gain amplifier we can configure it for a variety of *operations*: multiplication by a constant, summation, subtraction, differentiation, integration, and resistance conversion. This explains why it is called *operational*!

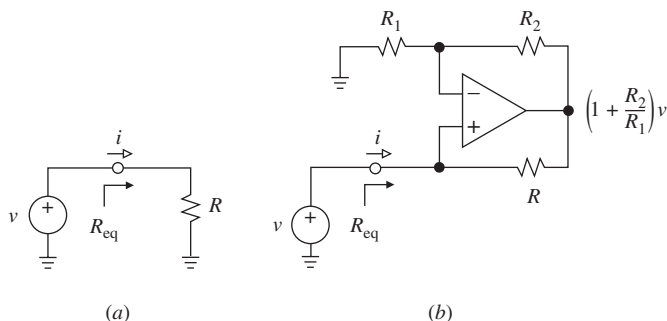


FIGURE 1.21

(a) Positive resistance: $R_{eq} = R$. (b) Negative-resistance converter: $R_{eq} = -(R_1/R_2)R$.

1.5
NEGATIVE FEEDBACK

Section 1.3 informally introduced the concept of negative feedback. Since the majority of op amp circuits employ this type of feedback, we shall now discuss it in a more systematic fashion.

Figure 1.22 shows the basic structure of a negative-feedback circuit. The arrows indicate signal flow, and the generic symbol x stands for either a voltage or a current signal. Besides the source and load, we identify the following basic blocks:

1. An amplifier called the *error amplifier*, which accepts a signal x_e called the *error signal*, and yields the *output signal*

$$x_o = a_e x_e \quad (1.37)$$

where a_e is called the *open-loop gain*.

2. A *feedback network*, which samples x_o and produces the *feedback signal*

$$x_f = b x_o \quad (1.38)$$

where b is the gain of the feedback network and is called the *feedback factor*.

3. A *summing network*, denoted as Σ , which sums the negative of x_f to the *input signal* x_i to yield the difference

$$x_e = x_i - x_f \quad (1.39)$$

The designation *negative feedback* stems from the fact that we are in effect feeding a portion b of x_o back to the error amplifier's input, where it is subtracted from x_i to yield a *reduced* signal x_e . Were it added instead, feedback would be positive. For reasons that will become clearer as we move along, negative feedback is also said to be *degenerative*, and positive feedback *regenerative*.

Substituting Eq. (1.38) into Eq. (1.39), and then into Eq. (1.37), we get

$$A = \frac{x_o}{x_i} = \frac{a_e}{1 + a_e b} \quad (1.40)$$

where A is called the *closed-loop gain* (not to be confused with the open-loop gain $a_e = x_o/x_e$). Note that for feedback to be negative we must have $a_e b > 0$. Consequently, A will be smaller than a_e by the amount $1 + a_e b$, which is called the *amount of feedback*. (Should there be no feedback, we would have $b = 0$ and $A \rightarrow a_e$, a situation referred to as open-loop operation.)

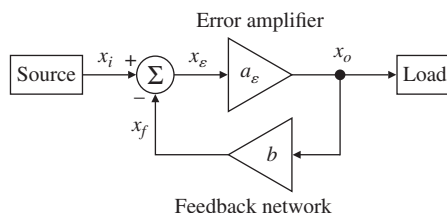


FIGURE 1.22
Block diagram of a negative-feedback system.

As a signal propagates around the loop consisting of the error amplifier, feedback network, and summer, it experiences an overall gain of $a_\varepsilon \times b \times (-1)$, or $-a_\varepsilon b$. Its negative shall be denoted as the *loop gain* L ,

$$L = a_\varepsilon b \quad (1.41)$$

This gain allows us to express Eq. (1.40) in the more insightful form $A = (1/b) \times L/(1 + L) = (1/b)/(1 + 1/L)$. Letting $L \rightarrow \infty$ yields the ideal situation

$$A_{\text{ideal}} = \lim_{L \rightarrow \infty} A = \frac{1}{b} \quad (1.42)$$

that is, A becomes independent of a_ε and is set exclusively by the feedback network, regardless of the error amplifier in use. By proper choice of the topology and components of the feedback network, we can tailor the circuit to a variety of different applications. For instance, specifying $b < 1$ so that $1/b > 1$, will cause x_o to be a *magnified* replica of x_i . Or, implementing the feedback network with reactive elements such as capacitors will yield a *frequency-dependent* circuit with the transfer function $H(s) = 1/b(s)$, where s is the complex frequency. Filters and oscillators are two popular examples.

Henceforth we shall express the closed-loop gain in the insightful form

$$A = A_{\text{ideal}} \frac{1}{1 + 1/L} \quad (1.43)$$

Rearranging as

$$A = A_{\text{ideal}} \left(1 - \frac{1}{1 + L} \right) \quad (1.44)$$

indicates that the *fractional departure* of the actual gain A from the ideal gain A_{ideal} is inversely proportional to the amount of feedback $1 + L$. This departure is more commonly expressed via the *gain error*

$$\text{GE}(\%) = 100 \frac{A - A_{\text{ideal}}}{A_{\text{ideal}}} = \frac{100}{1 + L} \quad (1.45)$$

EXAMPLE 1.7. (a) Find the loop gain needed for $\text{GE} \leq 0.1\%$. (b) Find a_ε to achieve $A = 50$ with the above GE. (c) What is the actual value of A ? How would you change b to make $A = 50.0$ exactly?

Solution.

- (a) Use Eq. (1.45) to impose $100/(1 + L) \leq 0.1$, or $L \geq 999$ (use $L \geq 10^3$).
- (b) Use Eq. (1.42) to impose $50 = 1/b$, or $b = 0.02$. Then, $L \geq 10^3 \Rightarrow a_\varepsilon b \geq 10^3 \Rightarrow a_\varepsilon \geq 10^3/0.02 = 5 \times 10^4$.
- (c) With $L = 10^3$ we have, by Eq. (1.43), $A = 49.95$. To make $A = 50.0$, use Eq. (1.40) to impose $50 = 5 \times 10^4/(1 + 5 \times 10^4 b)$, or $b = 0.01998$.

This example evidences the price for a low gain error, namely, the need to start out with $a_\varepsilon \gg A$ so that, once we close the loop around the error amplifier, we are in effect throwing away a good deal of open-loop gain, namely, the amount of feedback $1 + L$. It is also apparent that for a given a_ε , the smaller the desired gain A , the higher the feedback factor b , and thus the higher the loop gain L and the lower the gain error.

It is instructive to investigate the effect of negative feedback on the signals x_e and x_f . Writing $x_e = x_o/a_e = (Ax_i)/a_e = (A/a_e)x_i$, we get

$$x_e = \frac{x_i}{1 + L} \quad (1.46)$$

Moreover, writing $x_f = bx_o = b(Ax_i)$ and using Eq. (1.40), we get

$$x_f = \frac{x_i}{1 + 1/L} \quad (1.47)$$

As $L \rightarrow \infty$, the error signal x_e will approach zero, and the feedback signal x_f will track the input signal x_i . This forms the basis of the already-familiar virtual-short concept.

Gain Desensitivity

We wish to investigate how variations in the open-loop gain a_e affect the closed-loop gain A . Differentiating Eq. (1.40) with respect to a_e gives $dA/da_e = 1/(1 + a_e b)^2$. Substituting $1 + a_e b = a_e/A$ and rearranging,

$$\frac{dA}{A} = \frac{1}{1 + L} \frac{da_e}{a_e}$$

Replacing differentials with finite differences and multiplying both sides by 100, we approximate as

$$100 \frac{\Delta A}{A} \cong \frac{1}{1 + L} \left(100 \frac{\Delta a_e}{a_e} \right) \quad (1.48)$$

In words, the impact of a given percentage variation in a_e upon A is reduced by the amount of feedback $1 + L$. So long as L is sufficiently large, even a gross variation in a_e will cause an insignificant variation in A . We say that negative feedback desensitizes gain, this being the reason why the amount of feedback $1 + L$ is also called the *desensitivity factor*. The stabilization of A is highly desirable because the gain a_e of a real-life amplifier is ill-defined because of fabrication process variations, thermal drift, and aging.

What about the sensitivity of A to variations in b ? Again differentiating Eq. (1.40), but with respect to b , we get

$$100 \frac{\Delta A}{A} \cong 100 \frac{\Delta b}{b} \quad (1.49)$$

indicating that negative feedback does not stabilize A against variations in b . If we want a stable A , we need to implement the feedback network with components of adequate quality.

EXAMPLE 1.8. A negative-feedback circuit has $a_e = 10^5$ and $b = 10^{-3}$. (a) Estimate the percentage change in A brought about by a $\pm 10\%$ change in a . (b) Repeat if $b = 1$.

Solution.

- (a) The desensitivity factor is $1 + L = 1 + 10^5 \times 10^{-3} = 101$. Thus, a $\pm 10\%$ change in a_e will cause a percentage change in A 101 times as small; that is, A changes by $\pm 10/101 \cong \pm 0.1\%$.
- (b) Now the desensitivity increases to $1 + 10^5 \times 1 \cong 10^5$. The percentage change in A is now $\pm 10/10^5 = 0.0001\%$, or one part per million (1 ppm). We note that for a given a_e , the lower the value of A , the higher the desensitivity because $1 + L = a_e/A$.

VALUE {10*((exp(2E4*V(P,0))-1)/(exp(2E4*V(P,0))+1))}

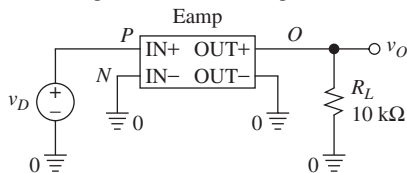


FIGURE 1.23
PSpice amplifier with a nonlinear voltage transfer curve (nonlinear VTC).

Nonlinear Distortion Reduction

A convenient way of visualizing the transfer characteristic of the error amplifier is by means of its *transfer curve*, that is, the plot of the output x_O versus the input x_E . The transfer curve of Eq. (1.37) is a straight line with slope a_ε . However, the transfer curve of a real-life amplifier is generally nonlinear. As an example, let us use the PSpice circuit of Fig. 1.23 to display the function

$$v_O = V_o \tanh \frac{v_E}{V_\varepsilon} = V_o \frac{\exp(2v_E/V_\varepsilon) - 1}{\exp(2v_E/V_\varepsilon) + 1} \quad (1.50)$$

which mimics qualitatively a real-life *voltage transfer curve* (VTC). Using $V_o = 10$ V and $V_\varepsilon = 100$ μ V, and directing PSpice to perform a dc sweep, we obtain the VTC of Fig. 1.24a, top. The voltage gain a_ε , visualized as the *slope* of the VTC,

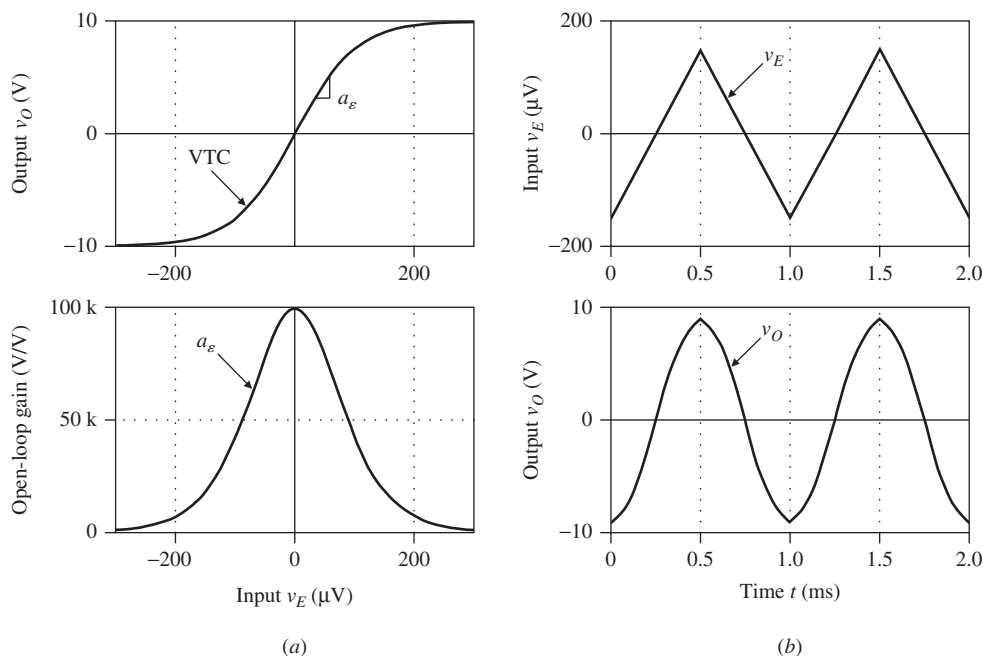


FIGURE 1.24
(a) Open-loop characteristics (VTC and gain a_ε) of the amplifier of Fig. 1.23. (b) The response v_O to a triangular input v_E .

VALUE {10*((exp(2E4*V(P,0))-1)/(exp(2E4*V(P,0))+1))}

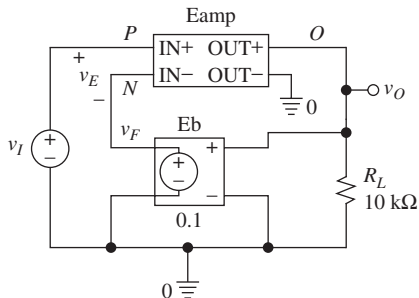


FIGURE 1.25

Applying negative feedback ($b = 0.1$) around the amplifier of Fig. 1.23.

is plotted by PSpice as the function $D(V(O))$. The result, shown at the bottom of Fig. 1.24a, indicates that a_e is maximum at the origin, but decreases as we move away from the origin, and eventually drops to zero at the periphery, where the VTC saturates at ± 10 V.

An obvious consequence of nonlinear gain is *distortion*, as depicted in Fig. 1.24b for the case of a triangular input. So long as we keep v_E sufficiently small (say, $\pm 10 \mu\text{V}$ or less), v_O will be a relatively undistorted, amplified version of v_E . However, increasing v_E increases the amount of output distortion, as exemplified in the picture. Increasing v_E further would lead to severe output *clipping* and thus even greater distortion.

What are we to make of an amplifier with the above nonlinearity/distortion shortcomings? Enter negative feedback, as depicted in Fig. 1.25, where we use the dependent source **EB** to sample the output-port voltage v_O , attenuate it to 1/10 to create the feedback signal v_F , and present v_F to the input port of the amplifier, where the latter subtracts it from v_I to create the error signal v_E (note that the operation of input summation is performed inherently by the amplifier itself). The results of the simulation, displayed in Fig. 1.26, reveal a dramatic linearization of the closed-loop VTC: the gain A is fairly constant ($A \cong 1/b = 10 \text{ V/V}$), and it is so over a much wider output range than the highly nonlinear gain a_e ; moreover, v_O is a fairly undistorted, amplified version of v_I . This, of course, holds over the voltage range for which a_e is sufficiently large to make the loop gain $L \gg 1$, as per Eqs. (1.42) and (1.43). As we approach the regions of open-loop saturation, where a_e falls, the linearizing effect of negative feedback no longer applies because of the lack of sufficient loop gain ($L = a_e b$) there, so A itself falls.

Given the highly nonlinear VTC of Fig. 1.24a, how does the amplifier manage to yield the undistorted waveform v_O of Fig. 1.26b? The answer is provided by the error signal v_E , also shown in Fig. 1.26b, which illustrates how negative feedback forces the amplifier to *predistort* (via the feedback network) its own input v_E in such a way as to yield the undistorted output v_O . In fact, it was the desire to reduce output distortion that motivated the negative-feedback concept in the first place!